1.Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux .

Ans. Ping Test Between Laptop and Phone

1. Connect **both the laptop and the friend's phone to the same Wi-Fi network**.
2. Find the **IP address of the phone** from its Wi-Fi/network settings. For example: 192.168.1.20.
3. On the laptop, open **Command Prompt**.
4. Type:
   ping 192.168.1.20
5. Press **Enter**.
6. If you see **"Reply from..."**, the laptop and phone can communicate with each other.
7. If you see **"Request timed out"**, the devices may not be allowed to communicate on the Wi-Fi network.

Example result:
Reply from 192.168.1.20: bytes=32 time=5ms

2.Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Traceroute Test

On **Windows**, follow these simple steps:

1. Open **Command Prompt**.
2. Type:

tracert www.youtube.com

3. Press **Enter**.
4. Wait for the test to finish.
5. Count the numbered lines (hops) shown in the result.

Example answer:

I ran the tracert www.youtube.com command on my laptop. It took **12 hops** to reach the destination.

3.Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Ans. Ping Test After Disconnecting the Device

1. First, connect both devices to the **same Wi-Fi**.
2. Find the phone's IP address, for example 192.168.1.20.
3. Disconnect the phone from Wi-Fi.
4. On the laptop, open **Command Prompt**.
5. Type:

ping 192.168.1.20

6. Press **Enter**.

Example result:
You may see **"Request timed out"** or **"Destination host unreachable."**

4.Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Ans. IP Address Table Using arp -a

On Windows, open Command Prompt and type:

arp -a

Press Enter. It will show the IP addresses of devices your laptop has recently communicated with on the Wi-Fi network.

You can make a simple table like this:

IP Address	Device / Person
192.168.1.1	Wi-Fi Router
192.168.1.2	My Laptop
192.168.1.3	Friend's Phone

IP Address	Device / Person
192.168.1.4	Smart TV
192.168.1.5	Brother's Phone

5.Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Include screenshots or command outputs if possible.

Ans. Network Setup and Connectivity Testing Report

Introduction

I set up my laptop and other devices on the same Wi-Fi network to test how they communicate with each other and with the internet. I used basic networking commands such as ping, tracert, and arp -a.

1. Connecting the Devices

First, I connected my laptop and a phone to the same home Wi-Fi network. I checked that both devices were connected properly and had IP addresses from the router.

2. Checking the IP Address

On my Windows laptop, I opened Command Prompt and entered:

ipconfig

This showed the laptop's IP address, subnet mask, and default gateway. The default gateway was the address of my Wi-Fi router.

3. Testing Local Connectivity

I found the phone's IP address and used the following command on my laptop:

ping [phone IP address]

The laptop received “Reply from...” messages, which showed that the laptop and phone could communicate over the Wi-Fi network.

Screenshot/Output: Add a screenshot of the successful ping result here.

4. Testing Internet Connectivity

I tested the connection to YouTube using:

ping www.youtube.com

The command returned replies, showing that my laptop had internet connectivity.

I then used:

tracert www.youtube.com

This showed the different network devices (hops) that the data passed through before reaching YouTube. The number of hops can change depending on the internet connection and network route.

Screenshot/Output: Add a screenshot of the tracert result here.

5. Checking Connected Devices

I used:

arp -a

This displayed IP addresses and MAC addresses of devices that my laptop had recently communicated with on the local network.

6. Observations and Issues

The ping test worked when both devices were connected to the same Wi-Fi. When I disconnected the phone and tried to ping it again, the result showed "Request timed out", because the phone was no longer available on the network. The traceroute test also showed that internet traffic passes through several hops before reaching the destination.

Conclusion

The network setup and connectivity tests were successful. The ping command helped check communication between devices, while tracert showed the route to an internet destination. The arp -a command helped identify devices on the

local network. These tests helped me understand how devices communicate through a Wi-Fi network.